AI-POWERED NETWORK INTRUSION DETECTION SYSTEM (AI-NIDS)

Comprehensive Product Requirements Document (PRD) & Technical Architecture Specification

Document Version: 2.2.0 (Release Candidate)	**Operating Mode:** Passive Zero-Payload Tap	**Target Budget:** ■0 (Zero Cloud Dependency)
Publication Date: October 26, 2026	**Deployment Target:** Localhost & Vercel Edge	**Test Suite:** 176 Automated Unit/Integration Tests

1. Executive Summary & Vision

The **AI-Powered Network Intrusion Detection System (AI-NIDS)** is an enterprise-grade, zero-cloud-cost, passive network threat detection platform. It provides Security Operations Center (SOC) teams with real-time bidirectional network flow reconstruction, supervised threat classification, calibrated unsupervised anomaly detection, sub-15ms Explainable AI (XAI) feature attribution, and a high-density single-page web console.

Core Architectural Principles:
• **■0 Budget Guarantee:** Strictly built on free and open-source software (FOSS). Zero commercial licenses, zero hosted databases.
• **100% Offline Capability:** Operates completely on-premises without internet access or phone-home telemetry.
• **Passive Traffic Monitoring:** Strictly non-destructive. Never performs active scans, packet drops, or automated firewall blocks.
• **Zero-Payload Privacy Rule:** All deep packet payloads are dropped at the socket layer; only transport metadata is inspected.

2. Dataset Architecture & Storage Locations

The detection engine is architected around the **CIC-IDS2017 Dataset** developed by the Canadian Institute for Cybersecurity (CIC) and the University of New Brunswick (UNB). It contains 2.83 million bidirectional flow records spanning 8 working-day capture sessions across benign traffic and 13 realistic attack vectors.

Exact Physical File Locations in Project:

Directory / File Path	Classification	Description & Operational Role
`data/sample_features.csv`	Bundled Test Data	10-flow verified datasheet adhering strictly to the 66 CIC-IDS2017 schema. Used for 1-click UI ingestion and CI testing.
`data/pcap/sample_network_traffic.pcap`	Synthetic PCAP	17-packet synthetic PCAP trace. Tests raw packet parsing, 5-tuple flow reconstruction, and ML classification.
`data/raw/`	Drop Directory	Target location for official raw CIC-IDS2017 multi-gigabyte CSV files (e.g. *Monday-WorkingHours.pcap_ISCX.csv*).
`data/processed/`	Metadata & Cache	Contains feature_metadata.json documenting column statistics, min/max bounds, and non-feature columns.
`data/alerts.db`	SQLite Datastore	Persistent local SQLite database storing alerts, 5-tuples, confidence scores, and analyst triage notes. (Mounted to /tmp/alerts.db on Vercel).

3. The 66 Canonical Flow Features & Attack Taxonomy

Network flows are characterized by 66 numeric attributes spanning flow durations, inter-arrival times (IAT), packet length moments, TCP control flags, and activity/idle cycles:

Category	Count	Sample Features Included	Forensic Significance
Timing & IAT	5	`flow_duration, flow_iat_mean, flow_iat_std, flow_iat_max, flow_iat_min`	Microsecond inter-packet pacing. Botnets and DoS scripts display low, fixed variance.
Packet Volumes	18	`total_fwd_packets, total_backward_packets, fwd_packet_length_mean, bwd_packet_length_std...`	Directional packet counts and payload sizes separating downloads from floods.
Transfer Rates	4	`flow_bytes_s, flow_packets_s, fwd_packets_s, bwd_packets_s`	Bandwidth velocity distinguishing volumetric DDoS from low-and-slow DoS.
TCP Control Flags	11	`fin_flag_count, syn_flag_count, rst_flag_count, psh_flag_count, ack_flag_count...`	TCP handshake flags. SYN spikes indicate port scans or SYN floods.
Segment & Window	8	`fwd_header_length, down_up_ratio, init_win_bytes_forward, init_win_bytes_backward...`	TCP window sizes and header ratios revealing OS fingerprinting tools.
Active / Idle	8	`active_mean, active_std, active_max, idle_mean, idle_std, idle_max...`	Transmission bursts vs dormant periods separating APTs from interactive traffic.

The 13 Attack Classes: benign, ddos, portscan, bot, dos goldeneye, dos hulk, dos slowhttptest, dos slowloris, ftp-patator, ssh-patator, web attack - brute force, infiltration, heartbleed.

4. Multi-Tier Machine Learning & Anomaly Engine

AI-NIDS employs a multi-tiered detection engine combining supervised classification, unsupervised anomaly scoring, and real-time surrogate explainability:

Model Layer	Algorithm & Config	Target Output	Operational Function
Binary Detector	Logistic Regression / Decision Tree (Recall-Prioritized)	is_malicious (0 / 1) Confidence score	High-recall gatekeeper. Missing an intrusion is treated as catastrophic.
Multiclass Classifier	Balanced Multi-Class Decision Tree & Random Forest	13 Attack Categories Class probability vector	Identifies specific threat category (DDoS, Bot, PortScan) with balanced class weights.
Anomaly Scorer	Isolation Forest (100 Trees) Contamination: 0.05	Normalized Anomaly Score [0.0..1.0] is_anomaly (0/1)	Detects zero-day anomalies and protocol deviations without prior signatures.
Explainable AI (XAI)	Decision Tree Surrogate Engine	Top-5 Feature Weights Logical Predicate Path	Sub-15ms local attribution explaining exact transport triggers to analysts.

5. Empirical Latency, Throughput & Memory Profiling

Benchmarked empirically on Darwin x86_64 (Python 3.11.9) with zero simulated timers:

Component / Metric	Median Latency	P95 Latency	P99 Latency	Max Throughput
Binary Classifier	10.73 ms	23.89 ms	63.20 ms	38,000 flows/sec (batch)
Multiclass Classifier	10.98 ms	33.01 ms	71.54 ms	40,475 flows/sec (batch)
Isolation Forest	30.18 ms	67.68 ms	99.28 ms	14,500 flows/sec (batch)
XAI Surrogate Engine	14.12 ms	60.74 ms	104.83 ms	7,200 flows/sec (batch)
PCAP Dissection	0.58 ms / pkt	1.20 ms / pkt	2.40 ms / pkt	1,727 packets/sec

Memory Footprint: Process Resident Set Size (RSS) is **218.58 MB**. Peak heap allocation during a 5,000-flow batch is **18.82 MB**.

6. Cloud Edge Deployment & Vercel WSGI Middleware

AI-NIDS is optimized for serverless edge deployment on Vercel (`api/index.py`). Two key challenges were solved:

1. **Overcoming Lambda 500 MB Size Limit:** Isolating research packages (Streamlit, Matplotlib) into requirements-dev.txt reduced the production bundle from 533.9 MB to **~220 MB**.
2. **VercelPathFixMiddleware Reverse-Proxy Solution:** In Vercel CLI 65+, internal rewrites forward requests with PATH_INFO = '/api/index'. AI-NIDS attaches a custom PEP 3333 WSGI middleware that intercepts HTTP_X_FORWARDED_URI and restores the true requested path (e.g. /api/download/sample-csv, /api/ingest/csv), preventing 405 Method Not Allowed errors and index.html download overrides.

7. REST API Endpoint Specification

Endpoint	Method	Payload / Params	Response Code & Description
`/`	GET	None	200 text/html — Serves SPA dashboard
`/api/health`	GET	None	200 JSON — Pipeline and DB health status
`/api/stats`	GET	None	200 JSON — Executive telemetry & KPI totals
`/api/alerts`	GET	status, class, limit	200 JSON — Filterable, paginated alert records
`/api/alerts/<id>/status`	POST	status, analyst_notes	200 JSON — Lifecycle status update
`/api/ingest/csv`	POST	file (multipart)	200 JSON — Batch CSV inference & alert persistence
`/api/ingest/pcap`	POST	file (multipart)	200 JSON — Raw PCAP packet dissection & inference
`/api/download/sample-csv`	GET	None	200 text/csv — Download verified 66-feature sample
`/api/export/csv`	GET	status, search	200 text/csv — Sanitized compliance report
`/api/alerts/clear`	POST	None	200 JSON — Reset local alert database to 0

8. Module-by-Module Codebase Anatomy

The AI-NIDS platform is composed of 14 core Python modules organized into a clean separation of concerns:

Module Path	Primary Responsibility	Key Classes & Functions
`src/config.py`	Central configuration, filesystem paths, hyperparameter bounds, and environment detection (VERCEL).	`PROJECT_ROOT, DATA_DIR, ALERTS_DB_PATH, NON_FEATURE_COLUMNS`
`src/data_loader.py`	Ingests multi-gigabyte raw CIC-IDS2017 CSV files, normalizes header casings, and isolates target labels.	`load_cicids_dataset(), stream_chunks(), generate_synthetic_dataset()`
`src/eda.py`	Statistical profiling, missing value audits, class imbalance calculation, and correlation matrices.	`run_eda(), plot_class_distributions(), generate_eda_report()`
`src/preprocess.py`	Mathematical sanitization, handling infinities, median imputation, and label encoding.	`clean_dataframe(), sanitize_infinities(), build_preprocessor()`
`src/feature_engineering.py`	Feature selection, zero-variance pruning, and standard/robust feature scaling.	`select_features(), scale_features(), save_feature_metadata()`
`src/train.py`	Supervised model training (Logistic Regression, Decision Trees, Random Forests) with balanced weighting.	`train_binary_models(), train_multiclass_models(), save_pipeline()`
`src/evaluate.py`	Classification reports, confusion matrices, ROC-AUC curves, and recall threshold validation.	`evaluate_model(), plot_confusion_matrix(), compute_metrics()`
`src/predict.py`	High-level unified inference engine orchestrating binary, multiclass, anomaly, and XAI pipelines.	`UnifiedPredictor, predict(), predict_single_flow()`
`src/anomaly.py`	Semi-supervised Isolation Forest anomaly scoring engine trained on benign traffic baselines.	`AnomalyDetector, fit_baseline(), score_samples()`
`src/explainability.py`	Sub-15ms decision-tree surrogate attribution replacing heavy SHAP computation.	`SurrogateExplainer, explain_flow(), extract_decision_path()`
`src/pcap_processor.py`	Passive raw packet parser, 5-tuple conversation tracker, and flow state machine.	`process_pcap(), PacketDissector, FlowAssembler`
`src/flow_features.py`	Real-time calculation of 66 statistical flow features from dissected packet arrival trains.	`extract_flow_features(), calculate_iat(), aggregate_flags()`
`src/alert_manager.py`	Persistent SQLite CRUD engine managing alert lifecycles, analyst notes, and compliance reports.	`AlertManager, create_alerts_from_df(), update_alert_status()`
`dashboard/server.py`	High-performance Flask 3.x REST API with VercelPathFixMiddleware reverse-proxy support.	`app, VercelPathFixMiddleware, ingest_csv(), download_sample_csv()`

9. Comparative Industry Matrix: AI-NIDS vs. Existing Solutions

Dimension	AI-NIDS	Snort 3.x	Suricata	Zeek (Bro)	Commercial Cloud NIDS
Cost	**■0 (100% Free)**	Free / FOSS	Free / FOSS	Free / FOSS	$5k–$50k / yr
Detection Basis	**Hybrid ML + Anomaly**	Signatures	Signatures	Behavioral Scripts	Black-Box Cloud AI
Zero-Day Resilience	**High (iForest)**	Poor (Needs hash)	Poor (Needs hash)	Medium	High
Zero-Payload Privacy	**Guaranteed**	Inspects Payload	Inspects Payload	Configurable	Cloud Ingestion Risk
Explainability	**<15ms XAI Surrogate**	Rule String	Rule String	Script Log Trace	None (Opaque)
Memory Overhead	**~218 MB RAM**	1–4 GB RAM	2–8 GB RAM	2–8 GB RAM	Agent Footprint
Deployment Simplicity	**Single Python App**	C++ Build Toolchain	Multi-threaded Daemon	Cluster Daemon	Cloud Agent Required

10. Verification Matrix & 176-Test Coverage

The codebase is validated by **176 automated test cases** across 11 test suites with 100% pass rate:
• tests/test_dashboard_server.py: REST API endpoints, sample CSV download, Vercel reverse proxy middleware, and status lifecycles (12 tests).
• tests/test_pcap_processor.py & test_phase5.py: Packet parsing, bidirectional flow reconstruction, and 66-feature schema conformance.
• tests/test_phase3.py & test_phase4.py: Supervised binary and multiclass models, recall thresholds, and ROC-AUC metrics.
• tests/test_phase6.py & test_phase7.py: SQLite alert persistence, sanitized CSV reports, Isolation Forest calibration, and XAI surrogates.
• tests/test_data_loader.py & test_validate_data.py: Missing value handling, non-numeric column removal, and target leakage prevention.

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